ε-δ-definition: $\forall \varepsilon > 0$, $\exists \delta > 0$, such that $|x - x_0| < \delta \Rightarrow |f(x) - A| < \varepsilon$

$\lim\limits_{x \to x_0} f(x) = A$ $\lim\limits_{x \to +\infty} f(x) = A$ $\lim\limits_{x \to -\infty} f(x) = A$ $\lim\limits_{|x| \to +\infty} f(x) = A$

$\lim\limits_{x \to x_0^-} f(x) = A$ $\lim\limits_{x \to x_0^+} f(x) = A$

Sandwich Theorem: $f(x) \le h(x) \le g(x)$ $\forall x \in (x_0 - \delta, x_0) \cup (x_0, x_0 + \delta)$

and $\lim\limits_{x \to x_0} f(x) = \lim\limits_{x \to x_0} g(x) = A \Rightarrow \lim\limits_{x \to x_0} h(x) = A$

Heine's theorem: $\lim\limits_{x \to x_0} f(x) = A \iff \lim\limits_{n \to +\infty} f(x_n) = A$, $\forall \{x_n\} \to x_0$

Infinitesimal quantity (无穷小量)

Defn: If $\lim\limits_{x \to x_0} f(x) = 0$, then we say f is an infinitesimal quantity as $x \to x_0$.

E.g. $\lim\limits_{x \to 0} x^2 = 0$ $\lim\limits_{x \to 0} \sin x = 0$ $\lim\limits_{x \to +\infty} \frac{1}{x} = 0$

Properties of infinitesimal quantity

① if $\lim\limits_{x \to x_0} f(x) = 0$, $\lim\limits_{x \to x_0} g(x) = 0$, $\Rightarrow \lim\limits_{x \to x_0} (f(x) \pm g(x)) = 0$

$\Rightarrow \lim\limits_{x \to x_0} f(x) \cdot g(x) = 0$

② if $\lim\limits_{x \to x_0} f(x) = 0$, $\lim\limits_{x \to x_0} g(x) = A$, $|A| < +\infty$ $\Rightarrow \lim\limits_{x \to x_0} f(x) \cdot g(x) = 0$

E.g: $\lim\limits_{x \to 0} x \cdot \sin \frac{1}{x} = 0$

③ if $\lim\limits_{x \to x_0} \frac{f(x)}{g(x)} = 0$, and $\lim\limits_{x \to x_0} f(x) = 0$, $\lim\limits_{x \to x_0} g(x) = 0$, then we say f is a **higher order** infinitesemal quantity than g, or, say

$f(x) = o(g(x))$ as $x \to x_0$

=

E.g. $\lim\limits_{x\to 0}\frac{f(x)}{g(x)}=\frac{x^2}{x}=0$ $\lim\limits_{x\to 0}\frac{1-\cos x}{\sin x}=\lim\limits_{x\to 0}\tan\frac{x}{2}=0$

④ If $\lim\limits_{x\to x_0} f(x)=0$, $\lim\limits_{x\to x_0} g(x)=0$, and $\lim\limits_{x\to x_0}\frac{f(x)}{g(x)}=C\neq 0$, then we say f and g have **same order** of infinitesemal quantity, or, say

$$f(x)=O(g(x)) \quad \text{as} \quad x\to x_0$$

E.g.: $\lim\limits_{x\to 0}\frac{\sin x}{x}=\lim\limits_{x\to 0}\frac{\cos x}{1}=1$ $\lim\limits_{x\to 0}\frac{x^2}{1-\cos x}=\lim\limits_{x\to 0}\frac{2x}{\sin x}=2$

⑤ In particular, if $\lim\limits_{x\to x_0}\frac{f(x)}{g(x)}=1$, we say f and g are **equivalent** infinitesemal quantity, or, say

$$f(x)\sim g(x) \quad \text{as} \quad x\to x_0$$

Commonly used equivalent infinitesemal quantity.

At $x_0=0$

$\sin x\sim x$ $\arcsin x\sim x$

$\tan x\sim x$ $\arctan x\sim x$

$\ln(x+1)\sim x$ $e^x-1\sim x$ $a^x-1\sim x\ln a$

$1-\cos x\sim\frac{1}{2}x^2$ $\sqrt[n]{1+x}-1\sim\frac{1}{n}x$

At $x_0=1$ $\ln x\sim x-1$

E.g.: ① $\lim\limits_{x\to 0}\frac{\arctan x}{\sin 4x}=\lim\limits_{x\to 0}\frac{x}{4x}=\frac{1}{4}$

② $\lim\limits_{x\to 0}\frac{\tan x-\sin x}{\sin(x^3)}=\lim\limits_{x\to 0}\frac{\frac{\sin x}{\cos x}-\sin x}{\sin(x^3)}$

$=\lim\limits_{x\to 0}\frac{\frac{\sin x}{\cos x}(1-\cos x)}{\sin x^3}=\lim\limits_{x\to 0}\frac{1}{\cos x}\frac{\sin x(1-\cos x)}{\sin x^3}$

$=\lim\limits_{x\to 0}\frac{x\cdot\frac{1}{2}x^2}{x^3}=\frac{1}{2}$

Infinite quantity (无穷大量)

Defn: If f is defined on $(x_0-\varepsilon, x_0)\cup(x_0, x_0+\varepsilon)$, and

$$\lim_{x\to x_0}|f(x)| = +\infty$$

then we say f is an infinite quantity as $x\to x_0$.

Properties: If $\lim_{x\to x_0} f(x) = +\infty$, $\lim_{x\to x_0} g(x) = +\infty$, $\lim_{x\to x_0} h(x) = A \neq 0$ and $|A| < +\infty$

then we have

① $\lim_{x\to x_0}(f(x)+g(x)) = +\infty$ $\quad\quad$ $\lim_{x\to x_0}(f(x)+h(x)) = +\infty$

② $\lim_{x\to x_0} f(x)\cdot g(x) = +\infty$ $\quad\quad$ $\lim_{x\to x_0} f(x)\cdot h(x) = +\infty$

Thm: ① Let f define on $(x_0-\varepsilon, x_0)\cup(x_0, x_0+\varepsilon)$ and $f\neq 0$ on this interval, if f is an infinitesemal quantity as $x\to x_0$, then $\frac{1}{f}$ is an infinite quantity as $x\to x_0$

② If a function g is an infinite quantity as $x\to x_0$, then $\frac{1}{g}$ is an infinitesemal quantity as $x\to x_0$.

E.x. ① $\lim_{x\to\frac{\pi}{2}} 2(\sin x - \cos x - x^2) = 2 - \frac{\pi^2}{2}$

② $\lim_{x\to 0} \frac{x^2-1}{2x^2-x-1} = 1$

③ $\lim_{x\to 1} \frac{x^2-1}{2x^2-x-1} = \lim_{x\to 1}\frac{(x+1)(x-1)}{(2x+1)(x-1)} = \lim_{x\to 1}\frac{x+1}{2x+1} = \frac{2}{3}$

⑥ $\lim_{x\to 0} \frac{(x-1)^3+(1-3x)}{x^2+2x^3} = \lim_{x\to 0}\frac{x^3-3x^2+3x-1+1-3x}{x^2+2x^3} = \lim_{x\to 0}\frac{x^3-3x^2}{x^2+2x^3}$

$= \lim_{x\to 0}\frac{x^2}{x^2}\cdot\frac{x-3}{1+2x} = -3$

⑤ $\lim_{x\to 4} \frac{\sqrt{1+2x}-3}{\sqrt{x}-2} = \lim_{x\to 4} \frac{(\sqrt{1+2x}-3)(\sqrt{1+2x}+3)}{(\sqrt{x}-2)(\sqrt{1+2x}+3)} = \lim_{x\to 4} \frac{2x+1-9}{(\sqrt{x}-2)(\sqrt{1+2x}+3)}$

$= \lim_{x\to 4} \frac{2(x-4)(\sqrt{x}+2)}{(\sqrt{x}-2)(\sqrt{x}+2)(\sqrt{1+2x}+3)} = \lim_{x\to 4} \frac{2(x-4)}{x-4} \frac{\sqrt{x}+2}{\sqrt{1+2x}+3} = \lim_{x\to 4} 2 \cdot \frac{2+2}{3+3} = \frac{4}{3}$

⑥ $\lim_{x\to 0} \frac{\sqrt{a^2+x}-a}{x} \quad (a>0) = \lim_{x\to 0} \frac{(\sqrt{a^2+x}-a)(\sqrt{a^2+x}+a)}{x(\sqrt{a^2+x}+a)} = \lim_{x\to 0} \frac{x}{x} \frac{1}{\sqrt{a^2+x}+a}$

$= \frac{1}{2a}$

⑦ $\lim_{x\to 1} \frac{x^n-1}{x^m-1} = \lim_{x\to 1} \frac{(x-1)(x^{n-1}+x^{n-2}+\cdots+x+1)}{(x-1)(x^{m-1}+x^{m-2}+\cdots+x+1)} = \frac{n}{m}$

2 Find the limits by using sandwich theorem

① $\lim_{x\to-\infty} \frac{x-\cos x}{x} = \lim_{x\to-\infty} 1-\frac{\cos x}{x} = 1 \qquad 1-\frac{1}{x} \le 1-\frac{\cos x}{x} \le 1+\frac{1}{x}$

② $\lim_{x\to+\infty} \frac{x\cdot\sin x}{x^2-4} = 0 \qquad \frac{-x}{x^2-4} \le \frac{x\cdot\sin x}{x^2-4} \le \frac{x}{x^2-4}$

③ $\lim_{x\to+\infty} \frac{[x]}{x} = 1 \qquad \frac{x-1}{x} < \frac{[x]}{x} < \frac{x+1}{x}$

3 Based on $\lim_{x\to 0} \frac{\sin x}{x} = 1$, find the following limits.

④ $\lim_{x\to 0} \frac{\sin 2x}{x} = \lim_{x\to 0} 2\cdot\frac{\sin 2x}{2x} = 2$

$\sin x^3 = \sin(x^3)$

$\sin^2 x = (\sin x)^2$

⑤ $\lim_{x\to 0} \frac{\sin x^3}{\sin^2 x} \quad \lim_{x\to 0} \frac{\sin x^3}{x^3} \cdot \frac{x^2}{\sin^2 x} \cdot x = 0$

⑥ $\lim_{x\to 0} \frac{\tan x}{x} = \lim_{x\to 0} \frac{\frac{\sin x}{\cos x}}{x} = \lim_{x\to 0} \frac{\sin x}{x} \cdot \frac{1}{\cos x} = 1$

⑦ $\lim_{x\to \frac{\pi}{2}} \frac{\cos x}{x-\frac{\pi}{2}}$ Let $y = x-\frac{\pi}{2} \quad x = y+\frac{\pi}{2}$

$\cos x = \cos\left(y+\frac{\pi}{2}\right) = -\sin y, \quad y\to 0 \text{ as } x\to\frac{\pi}{2}$

$= \lim_{y\to 0} \frac{\cos y+\frac{\pi}{2}}{y} = \lim_{y\to 0} \frac{-\sin y}{y} = 1$

(5) $\lim_{x\to 0} \frac{\tan x - \sin x}{x^3} = \lim_{x\to 0} \frac{\frac{\sin x}{\cos x} - \sin x}{x^3} = \lim_{x\to 0} \frac{\sin x - \sin x \cdot \cos x}{x^3 \cdot \cos x}$

$= \lim_{x\to 0} \frac{\sin x}{x}\left(\frac{1-\cos x}{x^2}\right)\frac{1}{\cos x} = 1 \cdot \frac{1}{2} \cdot 1 = \frac{1}{2}$

(6) $\lim_{x\to 0} \frac{\arctan x}{x}$

(7) $\lim_{x\to +\infty} x \sin\frac{1}{x}$

(8) $\lim_{x\to a} \frac{\sin^2 x - \sin^2 a}{x-a}$

6. Based on $\lim_{x\to+\infty}\left(1+\frac{1}{x}\right)^x = e$, find the following limits.

(1) $\lim_{x\to 0}(1+2x)^{\frac{1}{x}} = \lim_{x\to 0}\left[(1+2x)^{\frac{1}{2x}}\right]^2 = \lim_{|y|\to+\infty}\left[\left(1+\frac{1}{y}\right)^y\right]^2 = e^2$ $\quad y = \frac{1}{2x}$

(2) $\lim_{x\to 0}(1+\alpha x)^{\frac{1}{x}}$ $(\alpha > 0) = e^{\alpha}$

(3) $\lim_{x\to 0}(1+\tan x)^{\cot x}$ Let $y = \cot x$, $\frac{1}{y} = \tan x$ $|y|\to+\infty$ as $x\to 0$

(4) $\lim_{x\to 0}\left(\frac{1+x}{1-x}\right)^{\frac{1}{x}} = \lim_{x\to 0}\left(\frac{1-x+2x}{1-x}\right)^{\frac{1}{x}} = \lim_{x\to 0}\left(1+\frac{2x}{1-x}\right)^{\frac{1}{x}}$

$|y|\to+\infty$ as $x\to 0$

$\frac{1}{y} = \frac{2x}{1-x} = \frac{2}{\frac{1}{x}-1} \Rightarrow \frac{1}{x} - 1 = 2y \Rightarrow \frac{1}{x} = 2y+1$

$\Rightarrow \lim_{|y|\to+\infty}\left(1+\frac{1}{y}\right)^{2y+1} = \lim_{|y|\to+\infty}\left(1+\frac{1}{y}\right)^{2y}\cdot\left(1+\frac{1}{y}\right) = e^2 \cdot 1 = e^2$

(5) $\lim_{x\to+\infty}\left(\frac{3x+2}{3x-1}\right)^{2x+1}$

(6) $\lim_{x\to+\infty}\left(1+\frac{\alpha}{x}\right)^{\beta x}$

5. Using the equivalent infinitesemal property to find the limits.

at $x \to 0$

$\arctan x \sim x$

$\sqrt[n]{1+x} - 1 \sim \frac{1}{n}x$

$1 - \cos x \sim \frac{1}{2}x^2$

(1) $\lim\limits_{x\to\infty} \dfrac{x \arctan\frac{1}{x}}{x - \cos x} = \lim\limits_{x\to\infty} \dfrac{\arctan\frac{1}{x}}{\frac{1}{x}} \cdot \dfrac{1}{x - \cos x} = \lim\limits_{x\to\infty} \dfrac{1}{x - \cos x} = 0$

(2) $\lim\limits_{x\to 0} \dfrac{\sqrt{1+x^2} - 1}{1 - \cos x} = \lim\limits_{x\to 0} \dfrac{\frac{1}{2}x^2}{1 - \cos x} = 1$